Patch Min-Cut Planner

Skeptic review and remediation summary | lab_a | 2026-11-09T11:06:05+00:00

VERIFIED MODEL PLAN. Independent verification leaves zero modeled entry-to-crown paths. This is evidence-bounded, not proof that the real environment is safe.

Paths before	Paths after	Measured reduction	Patches	Confirmed / downgraded / rejected
1	0	100.0%	1	2 / 0 / 0

The optimizer selected 1 patch(es); independent verification reduced modeled paths from 1 to 0.

Selected patches and evidence

Patch / cost	Related finding and deterministic evidence
demo-db-fix Cost: 1	**demo-db:CVE-2099-90002:mariadb**: Confirmed for graph use only because the installed version matches an observed, reachable service, NVD CPE data marks the version affected, and NVD identifies AV:N. Scanner reports installed version 10.6.18 and fixed version 10.6.19.; Nmap observed mysql on db01:3306 (MariaDB 10.6.18).; Topology records 3 source(s) able to reach db01:3306.; NVD vulnerable CPE version range includes the installed version.; NVD CVSS vector identifies network attack vector (AV:N).

Rejected findings and reasons

None.

Finding verdicts

Finding	Verdict / confidence	Reason and evidence summary
demo-app:CVE-2099-90001:apache-httpd	**confirmed** **94%**	Confirmed for graph use only because the installed version matches an observed, reachable service, NVD CPE data marks the version affected, and NVD identifies AV:N. Scanner reports installed version 2.4.54 and fixed version 2.4.55.; Nmap observed http on app01:80 (Apache httpd 2.4.54).; Topology records 1 source(s) able to reach app01:80.; NVD vulnerable CPE version range includes the installed version.; NVD CVSS vector identifies network attack vector (AV:N).
demo-db:CVE-2099-90002:mariadb	**confirmed** **94%**	Confirmed for graph use only because the installed version matches an observed, reachable service, NVD CPE data marks the version affected, and NVD identifies AV:N. Scanner reports installed version 10.6.18 and fixed version 10.6.19.; Nmap observed mysql on db01:3306 (MariaDB 10.6.18).; Topology records 3 source(s) able to reach db01:3306.; NVD vulnerable CPE version range includes the installed version.; NVD CVSS vector identifies network attack vector (AV:N).

Limitations

- No exploits are run; scanner evidence and deterministic triage cannot prove exploitability.
- Package CVEs require exact observed service/version attribution; suggested ports are not proof. Missing NVD CPE or network-vector data is downgraded.
- Paths are simple and capped at 8 hops; topology/entry/crown labels may be incomplete. Patch cost is a heuristic. A verified plan cuts modeled paths only.
- SYNTHETIC DEMO FIXTURE ONLY. Not scanner output, not a real CVE, not real NVD data, and not an assessment of a real system. Uses explicitly fabricated future CVE identifiers to demonstrate the graph and patch planner.

No exploit was run. Review the accompanying report.html for the complete evidence ledger and limitations.